NUSRAT JAHAN LUNA

Bangamata Sheikh Fazilatunnesa Mujib Hall, Eden Mohila College, Dhaka-1205 · nusratjahan17.423@gmail.com · +880 13044 08422

EDUCATION

Eden Mohila College, Dhaka, Bangladesh. ***Oct '24***
BS (Hons.), Department of Chemistry

Viqarunnisa Noon College (Main Campus), Dhaka, Bangladesh. ***Dec '23***
Higher Secondary School Certificate (HSC), Science GPA 5.00/5.00

Matripith Government Girls' High School, Chandpur, Bangladesh. ***Jan '21***
Secondary School Certificate (SSC), Science GPA 5.00/5.00

TECHNICAL SKILS

Programming Language/Tools: C, C++, R, Python (Elementary)
Basic Software/Tools: MS Office, Slack, Google Workplace
Research Software/Tools: YASARA, Desmond, GROMACS, PyMOL, ChimeraX, Discovery Studio, GraphPad Prism, Gaussian etc

WORK EXPERIENCE

Research Assistant, Red Green Research Center ***Sept '25-Present***
- Worked on **Deep Eutectic Solvent (DES)** studies using computational chemistry and molecular simulation approaches.
- Conducted **molecular docking, molecular dynamics simulations, energy minimization, and trajectory analysis**.
- Used **YASARA, Desmond, and GROMACS** for MD simulations and system preparation.
- Applied **PyMOL and ChimeraX** for protein–ligand visualization, structural comparison, and figure preparation.
- Performed **quantum chemical calculations** using **Gaussian** for electronic structure analysis.
- Utilized **Discovery Studio** for interaction analysis and **GraphPad Prism** for statistical analysis and plotting.
- Assisted in data interpretation, result validation, and preparation of research materials.

Volunteer, National Outreach Coordinator (Bangladesh), International Astronomical Union ***Jan '25-Present***
- Organized the 3-day 17th Abdul Jabbar Astronomy Workshop, overseeing logistics, scheduling, and participant engagement.
- Led event operations in coordination with management and volunteer teams.
- Evaluated telescope options through technical and market analysis to support IAU-funded procurement.
- Translated astronomy outreach content from English to Bengali to expand local access.
- Created outreach-focused visuals for events, social media, and promotional campaigns.

Volunteer, Bangladesh Astronomical Association (BDAA) ***Jan '25-Present***
- Successfully completed a **TWO-DAY OBSERVATIONAL ASTRONOMY WORKSHOP**, a collaborative program between BDAA and the Russian House in Dhaka, enhancing practical astronomy skills.
- Served as an active member of the **AKASH PORJOBEKKHON TEAM**, contributing to public astronomy outreach initiatives.
- Assisted in organizing and facilitating a one-day sky observation event, guiding participants in viewing celestial objects (Moon, Venus, Mars) through an 8-inch reflective telescope.

Column Writer, The Daily Prothom Alo ***Mar '21-Jun '23***
- Wrote 7 articles in the Prothom Alo Gonit Ischool (single and with collaboration)
- Learn Microsoft Office (e.g. MS Word, Excel) software and formatting.

General Member, Eden Mohila College Research Society ***Nov '24-Present***
- Joined as a fresher, to build a community with the same interest field.
- Participated in some seminars and workshops with other members of the society.

Co-Head, Camp and Workshops, Deviation Scientific Society ***Mar '21-Jul '21***
- Managed 300+ students in Olympiad preparation camps and workshops.
- Organized 2 online camps (e.g. **AL-BIRUNI GEO CAMP**, **AL-KHWARIZMI AL-JABR CAMP**) with 700+ participants.
- Arranged 3+ contests for camp and workshop participants collaborating with DSS-Gonitzoggo.

Campus Ambassador, Gonitzoggo (gonitzoggo.com) ***Apr '22-Sept '22***
- Advertised some online workshops and helped to gather enthusiastic students.
- Learn LaTeX for adding new problems and formatting other problems.